

Online Retail: Customer Analytics & Segmentation

Final summary. Data: 541,909 transaction lines, 1 Dec 2010 - 9 Dec 2011, UK online retailer.

1. Data quality decisions

Issue	Decision	Business impact
5,268 exact duplicate lines	Removed	Avoids double-counting sales
9,288 cancellation lines	Kept as negative revenue (only 36.7% could be matched to an original purchase)	Revenue is net of returns
2,517 lines with UnitPrice <= 0	Excluded from the analysis dataset	Removes adjustments and stock write-offs
135,080 lines without CustomerID (3,710 whole invoices)	Kept for sales analysis, excluded from RFM and clustering	Guest invoices are 15.1% of net revenue

2. Key findings

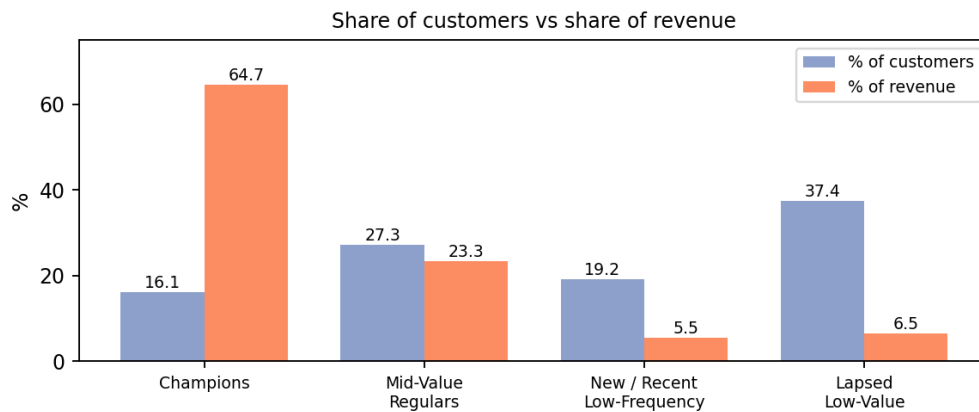
- **Revenue:** gross sales GBP 10.64M, returns GBP 0.89M (8.4% of gross), net revenue GBP 9.75M.
- **Products:** the top 20% of products generate about 79% of product revenue. Regency Cakestand 3 Tier is the top product (GBP 174k).
- **Customers:** the top 10% of customers generate 60.1% of net customer revenue.
- **Time:** monthly revenue is GBP 0.5M-0.75M until August, then rises to GBP 1.46M in November. December is incomplete (data ends 9 Dec). Thursday is the strongest weekday; activity peaks between 10:00 and 15:00.
- **Geography:** the UK generates 84% of revenue. International orders are larger (average GBP 845 vs GBP 500).
- **Baskets:** gross average order value GBP 533 (net GBP 410); 715 invoices exceed 1,000 items, so bulk buying exists. The two largest baskets were cancelled minutes later and are not counted as wholesale evidence.
- **Dead stock:** 22 of 918 high-volume products had no net sales in the last 90 days.

3. Method

RFM was built on 4,317 customers: Recency = days since last purchase, Frequency = unique purchase invoices, Monetary = net revenue. Each metric was scored 1-5 with quintiles (R: 5 = most recent). For clustering, R, F and M were log-transformed and standardized. K-Means with K = 4 was chosen using the Elbow method, silhouette scores (K=3: 0.338, K=4: 0.339; K=2 scores higher at 0.433 but is too coarse) and business interpretability. Hierarchical clustering (silhouette 0.268 vs 0.351 for K-Means on the same sample) and DBSCAN were used as checks.

4. Customer segments

Segment	Customers	Revenue share	Avg recency (days)	Avg orders	Avg net revenue (GBP)
Champions	697 (16.1%)	64.7%	12.1	13.9	7,694
Mid-Value Regulars	1,180 (27.3%)	23.3%	70.0	4.1	1,639
New / Recent Low-Frequency	827 (19.2%)	5.5%	17.4	2.2	552
Lapsed Low-Value	1,613 (37.4%)	6.5%	181.5	1.3	334



A separate check with RFM scores ($R \leq 2$, $F \geq 4$, $M \geq 4$) finds **173 high-value customers who stopped buying** (about 123 days since their last purchase). They hold GBP 382k of revenue (4.6%). DBSCAN also isolates 35 extreme accounts that generate 23.8% of revenue.

5. Recommendations

Segment	Action
Champions	Loyalty perks, early access to Q4 ranges, account-level service; manage the 35 whale accounts individually
Mid-Value Regulars	Cross-sell related products and bundles to raise order value and frequency
At-risk high-value (173)	Direct win-back offers now
New / Recent Low-Frequency	Welcome sequence with a second-purchase incentive
Lapsed Low-Value	Low-cost reactivation emails only; avoid heavy discounts (6.5% of revenue)

Operations: prepare stock and staff for Sept-Nov, schedule campaigns late morning on Tuesday/Thursday, review the 22 dead-stock products, and track returns by product and customer.

6. Limitations

- One year of data and an incomplete December: seasonality is suggested, not confirmed.
- Revenue is not profit (no cost data); guest invoices (15.1% of revenue) cannot be segmented.
- Silhouette of 0.34 means moderate separation: segment boundaries are soft. Findings are descriptive, not causal.